

REPUBLIC OF CAMEROON
Peace – Work – Fatherland

MINISTRY OF SECONDARY EDUCATION

INSPECTORATE GENERAL OF EDUCATION

Inspectorate of Pedagogy in charge of the
Teaching of Computer Science



REPUBLIQUE DU CAMEROUN
Paix – Travail – Patrie

MINISTRE DES ENSEIGNEMENTS SECONDAIRES

INSPECTION GENERALE DES ENSEIGNEMENTS

Inspection de Pédagogie chargée de
l'enseignement de l'informatique

NATIONAL HARMONISED LOWERSIXTH PROGRESSION SHEET FOR INFORMATION & COMMUNICATION TECHNOLOGY

SCHOOL YEAR.....

SCHOOL.....

WEEKLY WORKLOAD: 8 periods

TEACHER.....

GRADE.....

TEL.....

Term	Week	Module	Category of Action	Lesson Title	Objectives	Resources	Nature of Lesson		
							Th	Prac	Dig
First Term	1	Impacting society with digital technology	Applying Artificial Intelligence to improve on working and living standards	Lesson 1: The concept of artificial intelligence	- Define AI and explain its history and development . - Identify and describe the main types of AI. - Explain common application areas of AI.	AI SIMULATION AND AI TOOLS			
				Lesson 2: AI Ethics and Responsible Use	- Explain ethical issues associated with AI. - Examine the potential risks and benefits of AI. - Discuss principles for the responsible use of AI.				
				Lesson 3: AI Techniques and Intelligent Systems	- Describe common AI techniques and their applications. - Explain the characteristics of intelligent systems(smart school, smart city, smart health, smart home). - Compare AI techniques and intelligent systems based on their applications				
				Lesson 4: Machine Learning	- Define Machine Learning (ML) and explain its relationship with AI - Explain the basic principles of Machine Learning. - Build and test simple Machine Learning models using Teachable Machine.				
	2			Lesson 5: Developing AI Systems	- Describe the major stages involved in developing an AI system - Identify programming languages and tools used in AI development. - Explain why Python is widely used in AI development.				

				Lesson 6: Integration activities					
		Description of computer networking	Lesson 7: Fundamentals of computer Networks	- Describe a computer network - compare networked computers with stand-alone systems - Identify real-world examples of computer networks (e.g., home Wi-Fi, school labs, the internet) versus standalone computer setups.					
			Lesson 8: Network Complexity and Operational Challenges	- Identify and discuss the benefits and limitations of networks - Explain the complexity involved in setting up and maintaining data communication networks. - Explain the complexity of interoperability between heterogeneous systems.					
	3			Lesson 9: Integration activities					
		Classification of computer network	Lesson 10: Types of Networks by Coverage Area	- Describe types of networks (LAN, MAN, WAN, CAN, PAN) - Classify properly based on coverage a given network (e.g., home Wi-Fi setup, a university). - Compare the network types in terms of coverage area, typical speed, cost, and number of devices/users supported.					
			Lesson 11: Choosing the Right Network for an Organization	- Identify the networking needs of different types of organizations - Select the most appropriate network type for a given organizational scenario - Evaluate the key factors involved in choosing one network type over another for a specific organization's needs.					
			Lesson 12: Integration activities						
	4	Network Design Strategies	Lesson 13: network architecture (peer-to-peer and client/server)	- Explain network types based on the logical relationship between nodes. - Compare the advantages and disadvantages of each configuration. - Choose the appropriate network design for a given organisation					
		Configuration of network topology based on the Physical connection of computers	Lesson 14: Types of physical network topologies	- Describe the types of physical network topologies (mesh, star, ring, bus, tree). - Identify the physical layout and connection pattern of each topology. - Explain the advantages and disadvantages of each topology.					

				Lesson 15: Comparing and choosing Network Topologies	• Compare the different types of physical network topologies • Evaluate which topology suits specific organisational needs. • Propose a suitable network topology for a given organisation.				
				Lesson 16: Integration activities					
	5		Selection and configuration of network components	Lesson 17: Network equipment	• Explain the role of key network components (modem, hub, router, NOS...) • Differentiate between the various network components based on their functions • Identify the appropriate network components required for a computer network.				
				Evaluation No 1					
				Remediation No 1					
				Lesson 18: Setting Up a Computer Network	• Demonstrate the steps involved in setting up a basic computer network. • Identify the essential equipment and how they are interconnected. • Set up and test a peer-to-peer computer network.				
	6			Lesson 19: Selecting Equipment for Organizational Networks	• Select appropriate network equipment for specific organizational needs (e.g., a school) • Explain the factors to consider when selecting network components. • Plan and design a network setup suitable for an organization's size and purpose.				
				Lesson 20: Integration activities					
	Implementation of network security		Lesson 21: Intranet and Extranet	• Explain the concept of an intranet and an extranet • Explain the technologies and requirements used to implement intranets and extranets • Contrast the uses of intranets and extranets in organisations					
			Lesson 22: Virtual Private Networks (VPNs) and Privacy	• Describe the principles of a Virtual Private Network (VPN) • Explain how protocols are used to ensure privacy and confidentiality during data transmission; • Set up a secured VPN using firewalls and digital signatures					

	7			Lesson 23: Computer and Network Security	• Explain concepts related to computer and data security (confidentiality, integrity, availability) • Identify security measures used to protect computer systems (including user accounts, passwords, authentication, firewalls and security protocols; • Explain how digital signatures, encryption and other security measures help protect computer systems.				
				Lesson 24: Data Protection and Access Control	• Explain how encryption and access rights are used to protect computer data • Describe methods used to control access to data and prevent unauthorised use • Explain the importance of protecting data confidentiality, integrity and availability.				
				Lesson 25: Data Validation and Verification Techniques	• Explain and contrast the concepts of data validation and verification; • Describe validation techniques (e.g., range check, format check, length check... • Describe verification techniques such as visual check and double entry;				
				Lesson 26: Integration activities					
	8		Differentiating modes of data transmission	Lesson 27: Modes of data transmission	• • Explain logical topology methods (CSMA/CD, CSMA/CA) • • Describe different transmission modes (simplex, half-duplex, full duplex) • • Illustrate the direction of signal flow for a given data transmission scenario				
			Identifying different Transmission Types	Lesson 28: Broadband and Narrowband Transmission	• Define broadband and narrowband transmission; • Differentiate between broadband and narrowband transmissions; • Identify the appropriate transmission type used in a given communication situation.				
				Lesson 29: Types of data transmission	• Explain and contrast serial and parallel data transmission methods • Differentiate between synchronous and asynchronous transmission and identify their use in given situations; • Illustrate and differentiate between analog and digital signals.				
			Classification of transmission media	Lesson 30: Transmission media (guided and unguided)	• Describe guided (wired) transmission media (e.g., copper wire, coaxial cable, optical fibre...) • Describe unguided (wireless) transmission media (e.g., Bluetooth, microwave, Wi-Fi...) • Choose an appropriate transmission medium for a given context				

	9		Description of peripheral device control mechanism	Lesson 31: Peripheral device control mechanisms	• Explain the meaning and role of buffers, interrupts, interrupt priorities, polling and handshaking in controlling peripheral devices; • Differentiate between the various techniques used to control communication between the computer and peripheral devices; • Determine the technique used to control a device in a given network scenario				
			Using common protocols in data transmission	Lesson 32: Network Protocols and Their Roles in Data Transmission	• Explain the meaning and role of network protocols in data transmission • Discuss the functions of protocols (e.g., IP, TCP, HTTP...) in communication • Evaluate how TCP/IP and other protocols facilitate reliable and efficient data transmission.				
				Lesson 33: Protocol Suites and the OSI Reference Model	• Explain the role of protocol suites in network communication • Describe the seven layers of the OSI reference model and their functions • Relate network protocols to the appropriate layers of the OSI reference model.				
				Lesson 34: Integration activities					
				Evaluation No 2					
			Remediation No 2						
	10		Executing different correctness checking methods	Lesson 35: Error Detection and Data Transfer Checks	• Identify errors that may occur during data transmission • Describe data transfer checking mechanisms used to detect transmission errors • Explain parity checks as methods of error detection.				
			Internet technologies	Lesson 36: History and Development of the Internet	• Define the Internet and explain its purpose. • Recall major events in the history of the Internet. • Explain factors that led to growth of the internet. • Explain advantages and disadvantages of the internet.				
	11			Lesson 37: Internet Connectivity Technologies and Requirements	• Identify the technologies used to connect computers and networks to the Internet • Describe the role of devices in providing Internet connectivity. • Use core features of a browser.				

				Lesson 38: Social Networks and Collaborative Environments	• Define and Identify examples of social media and collaborative environments • Describe the positive and negative impacts of social networks and Internet use in the community; • Propose measures for the responsible and proper use of social media; • Demonstrate how collaborative platforms can be used for communication, learning and teamwork.				
			Using the internet	Lesson 39: Internet Services and Information Retrieval	• Describe the major services provided through the Internet • Create an e-mail account, attach a file to an e-mail and send it; • Research information on a particular topic using retrieval services.				
				Lesson 40: Internet Communication and Teleworking	• Explain the concepts of teleworking, telecommuting and videoconferencing • Explain the use of URLs and cookies when accessing web services • Set up and use a teleconferencing platform for online communication				
			Lesson 41: Integration activities						
	Learn with the assistance of computers		Lesson 42: Electronic services: e-commerce, e-health and e-government	• Define some major electronic services (e-commerce, e-health and e-government...) • Describe some major electronic services • Explain the advantages and disadvantages of using electronic services					
			Lesson 43: Applications, Benefits and Challenges of Electronic Services	• Compare the applications of e-commerce, e-health and e-government in everyday life; • Identify e-government platforms and explain their advantages and limits • Identify appropriate electronic services for solving common problems in the community.					
			Lesson 44: Computer-Assisted Learning (CAL) and E-Learning	• Describe Computer-Assisted Learning (CAL) and its role in the learning process • Identify different methods of course delivery (e.g., delivery through servers, CD-ROMs, e-books...) • Explain the advantages and limitations of CAL and e-learning;					
			Lesson 45: Integration activities						
	13		Develop simple web pages	Lesson 46: Developing simple web pages using HTML	• Identify the role of tags and attributes in a web page • Implement basic HTML tags to structure a web page • Create simple web pages using HTML and a text editor				

Second term			Apply common technologies, for web development	Lesson 47: Styling Web Pages with CSS	• Explain the purpose of Cascading Style Sheets (CSS) in web development • Write CSS code to modify the appearance of web pages • Apply CSS to improve the look and attraction of a simple web page.				
				Lesson 48: Creating interactive Web Pages with JavaScript	• Explain the role of JavaScript in web development • Write simple JavaScript code to process client-side and form data • Integrate HTML, CSS and JavaScript to create a simple interactive web page.				
	14		Host a website	Lesson 49: Website Hosting and Content Management Systems	• Explain the concepts of website hosting (domain names, public IP addresses, DNS...) • Exploit Content Management Systems to create and manage a website • Publish a simple website on the Internet using appropriate hosting and domain services.				
				Lesson 50: Integration activities					
	15	PRACTICAL PROBLEM SOLVING IN THE DIGITAL WORLD	Starting a business	Lesson 51: Identifying business needs solvable with IT	• Identify businesses related to basic ICT skills and explain customer needs • Discuss IT success stories and relate them to real business opportunities • Detect community needs that can be solved with information technology				
			Applying project management fundamentals to run sample projects	Lesson 52: Fundamentals of project management 1	• Define project management Explain its importance • Describe the activities involved at each stage of project management • Explain project management constraints such as scope, time and cost				
				Lesson 53: Fundamentals of project management 2	• Identify the roles and responsibilities of members of a project team. • Outline key factors that can contribute to project success or failure • Explain the concepts of tasks, critical task, Early Start (ES), Early Finish, slack, lag time, lead, dependency...				
				Evaluation No 3					
				Remediation No 3					
				Lesson 54: Project Planning and Work Breakdown Structure (WBS)	• Explain the purpose of project planning and the Work Breakdown Structure • Identify and organise the activities and tasks required to complete a project • Develop a Work Breakdown Structure for a simple project;				

	16			Lesson 55: Project Scheduling Using Gantt and PERT Charts	• Explain the purpose of Gantt charts and PERT charts in project scheduling; • Develop a simple project schedule using a Gantt chart; • Construct a PERT network to represent project activities and their dependencies; • Determine the duration of a project using appropriate scheduling techniques.				
				Lesson 56: Project Control, Critical Path and Slack Analysis	• Calculate ES, EF, LS and LF for activities in a project network; • Determine the critical path and identify critical and non-critical activities; • Analyse project progress and identify factors that may affect the successful completion of a project.				
				Lesson 57: Integration activities					
			Using formatting, editing features of Word processors for producing documents	Lesson 58: Producing documents using word processing features	• Apply formatting, editing, reviewing, and layout features to produce a professional document. • Use headings and styles to structure a long document and generate an automatic table of contents. • Use AI-assisted features to rewrite, summarize, or improve document content, and review the output for accuracy and relevance.				
	17	Carrying out statistical, financial etc. analysis, calculations and projections for education, businesses		Lesson 59: Formatting and managing spreadsheet data	• Apply cell formatting features (Add, delete, and resize rows and columns) • Format cell entries suitably for text, numbers, currency, percentages & dates • Implement conditional formatting to highlight important information in a dataset, using AI-assisted suggestions where appropriate. • Perform calculations using SUM, AVERAGE, PRODUCT, MAX, MIN.				
				Lesson 60: Cell referencing and basic formulae	• Distinguish between relative and absolute cell references. • Apply appropriate cell references to a cell or range of cells in a given problem. • Apply functions to perform financial and statistical analysis (IF, SUMIF, COUNTIF...)				
				Lesson 61: Advanced functions and charts	• Use the lookup function of a spreadsheet correctly. • Produce appropriate chart for a given scenario.				

				Lesson 62: Linking worksheets	• Explain how to link worksheets with formulae • Determine the appropriate spreadsheet formula for a given problem					
	18			Lesson 63: What-if-analysis	• Explain the concept and purpose of What-if analysis • Produce simple spreadsheet reports from analysed data • Use results of given formulae to make projections and informed decisions					
				Lesson 64: Integration activities						
			Producing slides that display illustrative and attractive presentations	Lesson 65: Producing slide presentations	• Explain the practices of a good presentation and the main features of presentation software; • Select appropriate presentation software and create slides using suitable designs and layouts; • Apply formatting techniques to produce clear, attractive and illustrative slides.					
				Lesson 66: Multimedia Presentations, Transitions and Animations	• Add and edit multimedia content using the features of presentation software; • Apply slide transitions and animations to create attention-grabbing presentations; • Apply basic project management principles to plan and deliver a presentation project.					
	Lesson 67: Integration activities									
	19		Using and RDBMS	Lesson 68: Creating tables and relationships	• Create tables and define constraints linked to tables using an RDBMS • Create relationships between tables.					
				Lesson 69: Queries 1	• Perform filtering and sorting of data using features of an RDBMS • Write simple queries using an RDBMS					
				Lesson 70: Queries 2	• Write queries using an RDBMS					
				20	Lesson 71: Reports and forms	• Create forms for data input using an RDBMS • Create reports using an RDBMS				
					Lesson 72: Integration activity					

	21	Cloud computing	Lesson 73: Introduction to cloud computing	• Explain the concept of cloud computing • Distinguish between local computing (on-premise hard drives, local RAM) and cloud computing (remote servers accessed via the Internet). • Describe core components (data centre, connectivity, virtualisation) of cloud computing. • Identify everyday consumer cloud tools (e.g., Google Drive, iCloud, Spotify, webmail). • Explain advantages and disadvantages of cloud computing					
			Lesson 74: Types of cloud computing services	• Compare the three fundamental cloud service models (Software as a Service, Platform as a Service, and Infrastructure as a Service) • Classify common digital platforms into their respective service model categories (e.g., Google Docs/Canva as SaaS, Firebase/Heroku as PaaS, AWS EC2/virtual machines as IaaS).					
			Lesson 75: Deployment Models, Security, and Cloud Ethics	• Differentiate between Public, Private, and Hybrid cloud deployment models based on cost, access control, and ownership. • Evaluate the basic security and privacy trade-offs of storing sensitive information on third-party remote servers. • Define the concept of shared responsibility between the cloud service provider and the end-user (e.g., vendor infrastructure security vs. user password hygiene and multi-factor authentication). • Debate a realistic case scenario involving data privacy, service outages, or vendor lock-in.					
			Lesson 76: Integration activity						
	22	Green computing	Evaluation No 4						
			Remediation No 4						
			Lesson 77: Introduction to green computing	• Define green computing and explain its importance. • Describe practices for using computer systems in an environmentally responsible and energy-efficient manner. • Identify practices that promote or do not promote green computing.					
			Lesson 78: Responsible Disposal	• Explain the environmental impact of improper disposal of computer systems and electronic waste. • Describe environmentally friendly methods of disposing of, reusing and recycling computer equipment. • Identify and execute simple projects that promote green computing in the school or community.					
			Lesson 79: Integration activity						

			Algorithms and programming	Lesson 80: Selection control structure	• Write algorithms that make use of selection control structures.				
	23			Lesson 81: Loop control structure	• Write algorithms that make use of loop control structures.				
				Lesson 82: Drawing flowcharts	• Represent algorithms into flowcharts.				
				Lesson 83: dry running	• Establish the correctness of an algorithm using dry running				
				Lesson 84: writing code 1	• Write programs in C or python that makes use of selection control structures.				
	24			Lesson 84: writing code 2	• Write programs in C or python that makes use of loop control structures.				
				Lesson 85: writing code 3	• Write programs in C or python that makes use of data structures.				
				Lesson 86: Integration activity					
				Evaluation No 5					
				Remediation No 5					
Third term	25	REVISION							
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